Particle Dynamics in Digital Black-Hole Spacetimes

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We study particle dynamics in a one-dimensional lattice Majorana-fermion model designed to emulate black-hole and white-hole spacetimes. By encoding the effective metric into position-dependent lattice couplings, we analyze the low-momentum dispersion relation and the associated effective light-cone structure. Using real-space Bogoliubov–de Gennes diagonalization and wave-packet time evolution, we show that packet trajectories agree with the corresponding geodesic motion for black-hole profiles. We also extract the surface gravity from the near-horizon growth of the wave-packet standard deviation and compare the numerical result with the analytical scaling. For the white-hole geometry, we show that wave-packet compression leads to finite-size reflection, with a stagnation time that scales logarithmically with the system size. We further clarify lattice-specific effects, including doubler-induced oscillations inside the black hole and partial transmission in the white-hole geometry. These results support lattice fermion models as practical platforms for quantum simulation of black-hole spacetime particle dynamics while clarifying the lattice artifacts that limit their range of validity.

I. INTRODUCTION

Quantum simulation provides a route to realizing and probing quantum theories by encoding their dynamics in controllable many-body systems [? ?]. A particularly compelling target is quantum field theory (QFT) in curved spacetime, which describes quantum fields propagating on prescribed gravitational backgrounds and predicts fundamental phenomena such as cosmological particle creation, the Unruh effect, and Hawking radiation [? ? ? ?]. These phenomena lie at the interface of quantum theory and gravitation, yet their direct observation remains exceptionally challenging. This has motivated the development of experimentally accessible platforms in which curved-spacetime QFTs and their characteristic dynamics can be investigated under controlled conditions.

A major route toward this goal has been analog gravity, pioneered by Unruh's demonstration that sound waves in a transonic fluid obey equations equivalent to those of a field near a black-hole horizon [?]. This idea has developed into a broad research program in which collective excitations in engineered media, such as phonons in Bose–Einstein condensates and electromagnetic waves in nonlinear optical systems, experience an emergent effective metric, typically within a hydrodynamic or long-wavelength regime [? ? ? ?]. Such platforms have provided powerful laboratory access to horizon physics and the associated wave-conversion phenomena. Their curved geometry, however, is an effective description of excitations supported by the host medium and is consequently tied to its microscopic dynamics and dispersive corrections, rather than arising from a direct discretization of a prescribed curved-spacetime QFT.

A conceptually distinct route is to discretize a prescribed curved-spacetime QFT itself so that the target theory is recovered in a controlled continuum limit. Quantum-walk and lattice approaches have pursued this direction by encoding geometry into discrete dynamics or spatially inhomogeneous local couplings [? ? ?]. In previous work by some of us, a general correspondence was established between Majorana QFTs in arbitrary $(1+1)$-dimensional curved spacetimes with periodic spatial topology and one-dimensional spin-$1/2$ systems, with the metric mapped directly onto space- and time-dependent exchange couplings and magnetic fields [?]. We refer to this framework, and more generally to such controlled lattice encodings of prescribed curved-spacetime QFTs, as *digital curved spacetimes.* Here, "digital" primarily denotes the spatial discretization of the field theory itself, rather than any specific gate-based protocol, while also reflecting the natural compatibility of the resulting local lattice Hamiltonians with digital quantum processors and other programmable quantum-simulation platforms [?]. The framework has subsequently been applied to thermal detector responses in an inflationary spacetime and extended to QFTs with spatial boundaries [? ?]. Unlike in analog-gravity platforms, the geometry is imposed through an explicit metric-to-coupling dictionary, and the target continuum theory is recovered systematically as the lattice spacing

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is reduced.

Black-hole spacetimes provide a particularly stringent test of such lattice encodings. Position-dependent hopping and spin models have been used to study transport, entanglement, and information transfer near nominal lattice horizons, with outward propagation interpreted in some works as stimulated Hawking radiation or information escape [? ? ? ?]. Before making such an interpretation, however, one must first verify that the lattice model reproduces the elementary causal dynamics of the target continuum QFT. A pre-existing one-particle wave packet initially inside a future horizon must propagate further into the black-hole interior rather than return to the exterior, whereas Hawking radiation, when present, is a distinct quantum-field-theoretic particle-production process. Establishing this basic correspondence and quantifying its finite-lattice corrections are therefore essential prerequisites for using lattice models to address genuinely quantum horizon phenomena.

In this work, we establish this correspondence for particle dynamics in digital black-hole spacetimes. Starting from a one-dimensional lattice Majorana model obtained through the metric-to-coupling correspondence [?], we demonstrate that localized wave packets quantitatively follow the corresponding continuum geodesics as the lattice spacing is reduced. In particular, wave packets initialized inside a future horizon propagate further into the black-hole interior, faithfully reproducing its one-way causal structure. We further show that the surface gravity can be extracted from the evolution of the wave-packet width, and clarify how finite-size effects, fermion doubling, and different lattice regularizations modify the approach to the continuum limit. By contrasting black-hole and white-hole geometries, we separate the universal causal dynamics of the target spacetime from regularization-dependent lattice effects. These results establish a controlled benchmark for digital curved spacetimes and provide a necessary foundation for their extension to intrinsically quantum horizon phenomena, including Hawking particle production and quantum-information transport, as well as to time-dependent geometries and implementations on programmable quantum processors.

II. DIGITAL BLACK-HOLE SPACETIME ON A LATTICE

We consider a massless Majorana field in a static $(1+1)$-dimensional curved spacetime described by the Painlevé–Gullstrand-type metric

$$ds^2 = -\big[1-\beta^2(x)\big]dt^2 - 2\beta(x)\,dt\,dx + dx^2. \tag{1}$$

This coordinate system is regular at the horizon and covers the exterior and interior regions within a single time slicing, allowing particle propagation across the horizon to be followed without encountering a coordinate singularity. A horizon is located at $x = x_{\mathrm{h}}$ satisfying

$$\big|\beta(x_{\mathrm{h}})\big| = 1. \tag{2}$$

The Hamiltonian of the massless Majorana field on this background is

$$\begin{aligned} H_{\mathrm{QFT}} = \int_0^\ell dx \Big[&-\frac{1}{2}\left(\phi^\dagger\partial_x\phi^\dagger - \phi\partial_x\phi\right) \\ &- \frac{i\beta(x)}{2}\left(\phi^\dagger\partial_x\phi + \phi\partial_x\phi^\dagger\right)\Big], \end{aligned} \tag{3}$$

where ℓ is the length of the spatial interval. The field operators satisfy the equal-time canonical anticommutation relation

$$\big\{\phi(t,x), \phi^\dagger(t,y)\big\} = \delta(x-y), \tag{4}$$

with all other anticommutators vanishing.

The field-to-spin dictionary developed in Ref. [?] is characterized by five space- and time-dependent functions. The three functions $\alpha(t,x)$, $\beta(t,x)$, and $\gamma(t,x)$ determine the spacetime metric, $\zeta(t,x)$ specifies a local gauge choice for the Majorana field, and $p(t,x)$ parametrizes a nontrivial freedom in the spin-lattice realization. To obtain the static Painlevé–Gullstrand geometry in Eq. (1), we set

$$\alpha(t,x) = \gamma(t,x) = 1, \qquad \beta(t,x) = \beta(x). \tag{5}$$

We also fix the gauge as $\zeta(t,x) = 0$ and, for simplicity, restrict $p(t,x)$ to a spatially and temporally uniform constant p. Under this specialization, the dictionary maps the Majorana QFT on the Painlevé–Gullstrand background onto an anisotropic XY chain with a bond-dependent Dzyaloshinskii–Moriya interaction and a transverse field:

$$\begin{aligned} H_{\mathrm{spin}} = &-\frac{1}{4\varepsilon}\sum_{j=0}^{L-2}\Big[(1+p)\sigma_j^x\sigma_{j+1}^x - (1-p)\sigma_j^y\sigma_{j+1}^y \\ &\qquad - \beta_{j+1/2}\left(\sigma_j^x\sigma_{j+1}^y - \sigma_j^y\sigma_{j+1}^x\right)\Big] \\ &- \frac{p}{2\varepsilon}\sum_{j=0}^{L-1}\sigma_j^z. \end{aligned} \tag{6}$$

The shift function $\beta(x)$, which specifies the spacetime geometry, is encoded in the bond-dependent Dzyaloshinskii–Moriya coupling. Here, we evaluate the shift function at the center of each bond,

$$\beta_{j+\frac{1}{2}} \equiv \beta\left(x_{j+\frac{1}{2}}\right), \qquad x_{j+\frac{1}{2}} = \varepsilon\left(j+\frac{1}{2}\right). \tag{7}$$

This bond-centered convention aligns the sampling point of $\beta(x)$ with the spatial location of the Dzyaloshinskii–Moriya coupling that it determines. It differs from the site-centered convention $\beta_j = \beta(x_j)$ adopted in Ref. [?] only at finite lattice spacing, and the two prescriptions yield the same continuum QFT as $\varepsilon \to 0$.

For the analytical and numerical treatment below, we rewrite the spin Hamiltonian in terms of Jordan–Wigner fermions. For a finite open chain, this gives

$$H = -\frac{1}{2\varepsilon}\sum_{j=0}^{L-2}\Bigg[c_{j+1}c_j + c_j^\dagger c_{j+1}^\dagger + p\left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j\right) + i\beta_{j+\frac{1}{2}}\left(c_j^\dagger c_{j+1} - c_{j+1}^\dagger c_j\right)\Bigg] - \frac{p}{2\varepsilon}\sum_{j=0}^{L-1}\left(1-2c_j^\dagger c_j\right). \tag{8}$$

The operators satisfy

$$\{c_j, c_{j'}^\dagger\} = \delta_{jj'}, \tag{9}$$

with all other anticommutators vanishing.

As established in Ref. [?], this Jordan–Wigner representation is related to the continuum Majorana field through the spatial discretization

$$c_j = \sqrt{\varepsilon}\,\phi(x_j), \qquad x_j = \varepsilon j, \tag{10}$$

with the continuum theory recovered as $\varepsilon \to 0$ at fixed physical length $\ell = \varepsilon L$. Thus, Eq. (8) is simultaneously the exact fermionic representation of the local spin simulator and a controlled lattice realization of the target curved-spacetime QFT.

As shown in Ref. [?], the function $p(t,x)$ has two distinct roles in the field-to-spin dictionary. First, it participates in the realization of spatial boundary conditions in a finite open chain. Second, it controls the ultraviolet structure of the bulk lattice theory, including the presence and momentum-space structure of fermion doublers, without modifying the target spacetime geometry or continuum QFT. In the present work, the horizon and the initial wave packets are placed sufficiently far from the physical ends of the chain that boundary effects are negligible over the time scales considered. We therefore restrict $p(t,x)$ to the constant choices

$$p = 0 \qquad \text{and} \qquad p = 1, \tag{11}$$

and focus on its second role, namely the regularization-dependent bulk dynamics.

A. Local dispersion and causal velocities

To identify the bulk modes relevant to the wave-packet dynamics, we examine the local dispersion relation of the lattice model. Since $\beta(x)$ varies in space, momentum is not a good quantum number for the full open chain. We therefore consider an auxiliary homogeneous chain of L' sites in which $\beta(x)$ is replaced by a constant local value β and periodic boundary conditions are imposed. Here, L' denotes the number of sites in this auxiliary homogeneous system and is distinct from the size L of the open chain used in the real-space simulations.

Using the Fourier transformation

$$c_j = \frac{1}{\sqrt{L'}}\sum_k e^{ikj}c_k, \qquad k = \frac{2\pi n}{L'}, \tag{12}$$

where $k \in [-\pi, \pi)$ is the dimensionless lattice momentum, the homogeneous Hamiltonian can be written in the Bogoliubov–de Gennes form

$$H = \frac{1}{2}\sum_k \Psi_k^\dagger \mathcal{H}(k)\Psi_k + \text{const.}, \tag{13}$$

where

$$\Psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}, \tag{14}$$

and

$$\mathcal{H}(k) = \frac{1}{\epsilon}\begin{pmatrix} \beta\sin k + p(1-\cos k) & -i\sin k \\ i\sin k & \beta\sin k - p(1-\cos k) \end{pmatrix}. \tag{15}$$

The two eigenvalue branches are

$$E_\pm(k) = \frac{1}{\epsilon}\left[\beta\sin k \pm \left\{\sin^2 k + p^2(1-\cos k)^2\right\}^{1/2}\right]. \tag{16}$$

To take the continuum limit, we introduce the physical momentum

$$q = \frac{k}{\epsilon} \tag{17}$$

and let $\epsilon \to 0$ at fixed q, so that $k = \epsilon q \to 0$. The dispersion then becomes

$$E_\pm(\epsilon q) = \beta q \pm |q| + O\big(\epsilon^2|q|^3\big). \tag{18}$$

For the positive-momentum sector, $q > 0$, this reduces to

$$E_\pm(\epsilon q) = (\beta \pm 1)q + O\big(\epsilon^2 q^3\big). \tag{19}$$

The corresponding physical group velocities are therefore

$$v_\pm = \frac{\partial E_\pm}{\partial q} = \epsilon\frac{\partial E_\pm}{\partial k} \longrightarrow \beta \pm 1 \qquad (\epsilon \to 0). \tag{20}$$

These coincide with the two null characteristic velocities of the Painlevé–Gullstrand metric,

$$\frac{dx}{dt} = \beta(x) \pm 1, \tag{21}$$

obtained by setting $ds^2 = 0$.

The corresponding local dispersion relations are shown in Figs. 1(a) and 1(b) for $p = 0$ and $p = 1$, respectively. The three columns correspond to $\beta = 0$, 1, and 2. For $\beta = 0$, the two low-momentum branches have opposite group velocities, $v_\pm = \pm 1$. At $\beta = 1$, one of the characteristic velocities vanishes, signaling the formation of

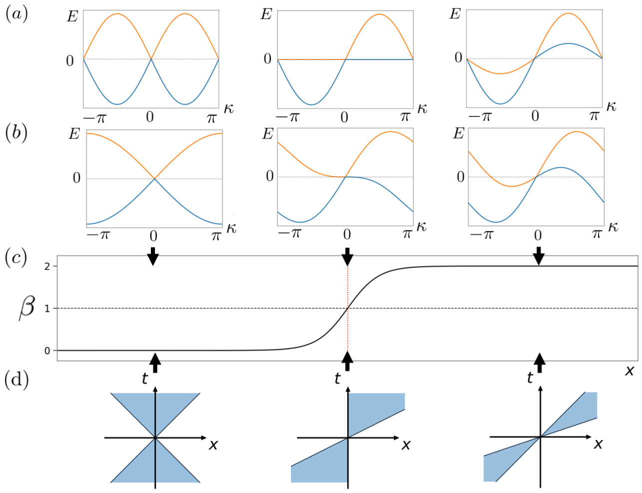

FIG. 1. Local dispersion relations for (a) $p = 0$ and (b) $p = 1$ at $\beta = 0$, 1, and 2, shown from left to right. (c) An example of the spatial profile $\beta(x)$ used in the real-space analysis. The dashed line marks $\beta = 1$, where the horizon is located. (d) Sketches of the corresponding local light cones determined by the low-momentum group velocities. Inside the horizon, where $\beta > 1$, both characteristic velocities are positive and point toward the black-hole interior.

a horizon. For $\beta > 1$, both low-momentum velocities become positive, and hence both branches propagate toward increasing x.

Although the two choices of p reproduce the same low-momentum dispersion, their finite-momentum structures are qualitatively different. For $p = 0$, additional zero-energy modes occur at the Brillouin-zone edges $k = \pm\pi$, corresponding to conventional fermion doublers. Their physical momenta are of order $|q| \sim \pi/\epsilon$, even though their excitation energies remain low. For $p > 0$, the term $p(1 - \cos k)$ lifts the zero-energy modes at the Brillouin-zone edges. Its momentum dependence is analogous to that of the Wilson term introduced to remove fermion doublers in lattice gauge theories [**?**]. In the black-hole interior $|\beta| > 1$, however, additional zero-energy crossings necessarily appear at finite lattice momenta $k = \pm k_d$, where

$$k_d = \arccos\left(\frac{p^2 - \beta^2 + 1}{p^2 + \beta^2 - 1}\right). \tag{22}$$

Indeed, for $\beta > 1$, the lower branch has positive energy immediately to the right of $k = 0$ but negative energy at $k = \pi$ for any $p \neq 0$, so the periodic lattice dispersion must cross zero at an intermediate momentum. The Wilson-like term therefore removes the conventional zone-edge doubler outside the horizon, but inside the overtilted region it shifts the unavoidable zero-energy lattice modes away from the Brillouin-zone edge rather than eliminating them.

Thus, although the two regularizations reproduce the same continuum causal velocities, they possess distinct finite-momentum modes that can affect the dynamics at finite lattice spacing. We now turn to real-space wave-packet dynamics to examine how these continuum and lattice modes manifest themselves in the inhomogeneous black-hole geometry.

B. Real-space dynamics and wave-packet preparation

The preceding analysis provides a local description of the particle dynamics by treating $\beta(x)$ as approximately constant. We now restore the full spatial dependence of the couplings and study the finite open-chain Hamiltonian directly in real space.

The quadratic fermion Hamiltonian in Eq. (8) can be written in the Bogoliubov–de Gennes form

$$H = \frac{1}{2}\mathbf{\Psi}^\dagger \mathcal{H}_{\rm BdG}\mathbf{\Psi} + \text{const.}, \tag{23}$$

where

$$\mathbf{\Psi} = \left(c_0, \ldots, c_{L-1}, c_0^\dagger, \ldots, c_{L-1}^\dagger\right)^{\mathsf{T}}. \tag{24}$$

We diagonalize $\mathcal{H}_{\rm BdG}$ by a fermionic Bogoliubov transformation

$$\mathbf{\Psi} = U\mathbf{\Gamma}, \qquad \mathbf{\Gamma} = \left(\gamma_1, \ldots, \gamma_L, \gamma_1^\dagger, \ldots, \gamma_L^\dagger\right)^{\mathsf{T}}, \tag{25}$$

which brings the Hamiltonian to the diagonal form

$$H = \sum_{n=1}^{L} E_n \gamma_n^\dagger \gamma_n + \text{const.}, \qquad E_n \geq 0. \tag{26}$$

The BdG vacuum is defined by

$$\gamma_n \left|\text{vac}\right\rangle = 0, \qquad n = 1, \ldots, L. \tag{27}$$

Since the quasiparticle occupation numbers $\gamma_n^\dagger\gamma_n$ are conserved, the dynamics of an initial one-quasiparticle state can be evaluated within the subspace spanned by

$$\left|n\right\rangle = \gamma_n^\dagger \left|\text{vac}\right\rangle, \qquad n = 1, \ldots, L. \tag{28}$$

To distinguish the two continuum null branches, we introduce the Hermitian operators $a_j^\pm$ through

$$c_j = \frac{1}{2}\left[a_j^+ - a_j^- - i\left(a_j^+ + a_j^-\right)\right], \tag{29a}$$

$$c_j^\dagger = \frac{1}{2}\left[a_j^+ - a_j^- + i\left(a_j^+ + a_j^-\right)\right]. \tag{29b}$$

They satisfy

$$\left\{a_j^s, a_{j'}^{s'}\right\} = \delta_{jj'}\delta_{ss'}, \qquad s, s' = \pm. \tag{30}$$

In terms of these operators, the lattice Hamiltonian becomes

$$\begin{aligned} H = & -\frac{i}{2\varepsilon}\sum_{j=0}^{L-2}\Big[\Big(1+\beta_{j+\frac{1}{2}}\Big)a_j^+a_{j+1}^+ \\ & \qquad -\Big(1-\beta_{j+\frac{1}{2}}\Big)a_j^-a_{j+1}^- \\ & \qquad +p\left(a_j^-a_{j+1}^+-a_j^+a_{j+1}^-\right)\Big] \\ & -\frac{ip}{\varepsilon}\sum_{j=0}^{L-1}a_j^+a_j^-. \end{aligned} \tag{31}$$

For $p=0$, the a^+ and a^- components are completely decoupled. A nonzero p mixes the two components through the Wilson-like regularization term.

The interpretation of the two components follows from their Heisenberg equations of motion. At a bulk site, they read

$$\begin{aligned} i\frac{\partial a_j^+}{\partial t} = & -\frac{i}{2\varepsilon}\Big[\Big(1+\beta_{j+\frac{1}{2}}\Big)a_{j+1}^+ \\ & \qquad -\Big(1+\beta_{j-\frac{1}{2}}\Big)a_{j-1}^+\Big] \\ & -\frac{ip}{2\varepsilon}\left(2a_j^--a_{j+1}^--a_{j-1}^-\right), \end{aligned} \tag{32a}$$

$$\begin{aligned} i\frac{\partial a_j^-}{\partial t} = & \frac{i}{2\varepsilon}\Big[\Big(1-\beta_{j+\frac{1}{2}}\Big)a_{j+1}^- \\ & \qquad -\Big(1-\beta_{j-\frac{1}{2}}\Big)a_{j-1}^-\Big] \\ & +\frac{ip}{2\varepsilon}\left(2a_j^+-a_{j+1}^+-a_{j-1}^+\right). \end{aligned} \tag{32b}$$

The p-dependent terms take the form of a discrete second derivative and are suppressed for smooth low-momentum modes in the continuum limit.

For smooth low-momentum modes, the p-dependent mixing terms are of order ε and vanish in the continuum limit. To leading order, the two components therefore obey independent transport equations,

$$\left[\partial_t+v_s(x)\partial_x+\frac{1}{2}\partial_x v_s(x)\right]a^s(t,x)=0, \qquad s=\pm, \tag{33}$$

with characteristic velocities

$$v_+(x)=\beta(x)+1, \qquad v_-(x)=\beta(x)-1. \tag{34}$$

For $p=0$, the two components are exactly decoupled at any lattice spacing. For $p\neq 0$, the mixing is suppressed for sufficiently smooth low-momentum modes and vanishes systematically toward the continuum limit. These velocities agree with the two null characteristic velocities obtained from the continuum metric. The a^+ and a^- components therefore have horizons at $\beta(x_\mathrm{h})=-1$ and $\beta(x_\mathrm{h})=1$, respectively. In the geometry considered below, $\beta(x)$ increases through 1, so that the black-hole horizon is associated with the a^- branch.

We prepare an initial wave packet by applying a Gaussian superposition of either a_j^+ or a_j^- to the BdG vacuum:

$$|\psi_s(0)\rangle=\mathcal{N}_s\sum_{j=0}^{L-1}G_j a_j^s\,|\mathrm{vac}\rangle\,, \qquad s=\pm, \tag{35}$$

where

$$G_j=\exp\left[-\frac{(j-j_0)^2}{2\sigma^2}\right]. \tag{36}$$

Here, j_0 and σ denote the center and width parameter of the Gaussian envelope, respectively. The normalization constant $\mathcal{N}_s$ is chosen such that

$$\langle\psi_s(0)|\psi_s(0)\rangle=1. \tag{37}$$

For $\sigma\gg 1$, the momentum distribution is concentrated near $k=0$, where the lattice dispersion approaches that of the target continuum QFT. The use of a_j^+ or a_j^- therefore selectively prepares a wave packet associated with the corresponding continuum null branch. For $p=0$, this separation is exact at any lattice spacing, while for $p\neq 0$ the mixing between the two components remains strongly suppressed for sufficiently smooth low-momentum wave packets.

Expanding the initial state in the one-quasiparticle eigenbasis, its time evolution is given by

$$|\psi_s(t)\rangle=\sum_{n=1}^{L}e^{-iE_nt}\,|n\rangle\,\langle n|\psi_s(0)\rangle\,, \tag{38}$$

where an overall phase associated with the vacuum energy has been omitted. We compare the resulting real-space propagation with the corresponding continuum null trajectory,

$$\frac{dx_s}{dt}=\beta(x_s)+s, \qquad x_s(0)=\varepsilon j_0, \qquad s=\pm 1. \tag{39}$$

To resolve the two propagation branches locally, we decompose the Hamiltonian density as

$$H=\varepsilon\sum_{j=0}^{L-1}\mathcal{H}_j, \qquad \mathcal{H}_j=\mathcal{H}_j^++\mathcal{H}_j^-+\mathcal{H}_j^{\mathrm{int}}, \tag{40}$$

where

$$\begin{aligned} \mathcal{H}_j^+ = & -\frac{i}{4\varepsilon^2}\Big[\Big(1+\beta_{j+\frac{1}{2}}\Big)a_j^+a_{j+1}^+ \\ & \qquad +\Big(1+\beta_{j-\frac{1}{2}}\Big)a_{j-1}^+a_j^+\Big], \end{aligned} \tag{41}$$

$$\begin{aligned} \mathcal{H}_j^- = & \frac{i}{4\varepsilon^2}\Big[\Big(1-\beta_{j+\frac{1}{2}}\Big)a_j^-a_{j+1}^- \\ & \qquad +\Big(1-\beta_{j-\frac{1}{2}}\Big)a_{j-1}^-a_j^-\Big], \end{aligned} \tag{42}$$

and

$$\begin{aligned} \mathcal{H}_j^{\mathrm{int}} = & -\frac{ip}{4\varepsilon^2}\Big[a_j^-a_{j+1}^+-a_j^+a_{j+1}^- \\ & \qquad +a_{j-1}^-a_j^+-a_{j-1}^+a_j^-+4a_j^+a_j^-\Big]. \end{aligned} \tag{43}$$

At the physical ends of the chain, terms containing operators outside the interval are omitted.

For a wave packet initially prepared from the a^s component, we monitor the corresponding branch-resolved, vacuum-subtracted energy density,

$$\delta\mathcal{E}_{j,s}(t) = \langle\psi_s(t)|\, \mathcal{H}_j^s\, |\psi_s(t)\rangle - \langle\text{vac}|\, \mathcal{H}_j^s\, |\text{vac}\rangle\,. \quad (44)$$

Thus, for an initial a^+ (a^-) wave packet, we plot the $\mathcal{H}_j^+$ ($\mathcal{H}_j^-$) contribution. This branch-resolved quantity isolates the component associated with the corresponding continuum null direction. For $p \neq 0$, it should be regarded as a branch-resolved diagnostic rather than the full local energy density, since the p-dependent terms can transfer weight between the two components at finite lattice spacing. Such mixing is suppressed for sufficiently smooth low-momentum wave packets and vanishes systematically in the continuum limit.

Our interpretation of the horizon dynamics focuses on the interval before appreciable wave-packet weight reaches the physical ends of the chain. Some plotted intervals extend to later times in order to display lattice collapse, reflection, or spectral evolution; these late-time portions are not used for quantitative comparison with boundary-free continuum trajectories.

III. WAVE-PACKET DYNAMICS ACROSS BLACK- AND WHITE-HOLE HORIZONS

We now study the real-space dynamics of the wave packets introduced in Sec. II. We first examine their propagation in a black-hole geometry and compare the lattice dynamics with the corresponding continuum null trajectories. We then investigate finite-spacing effects associated with additional finite-momentum lattice modes, extract the surface gravity from near-horizon wave-packet broadening, and finally contrast the results with wave-packet dynamics in a white-hole geometry.

Throughout this section, we use the shift profile

$$\beta(x) = A\tanh\left[B(x - x_0)\right] + C, \quad (45)$$

where x_0 is the inflection point of the profile and A, B, and C are real parameters. For the branch with characteristic velocity

$$v_s(x) = \beta(x) + s, \qquad s = \pm 1, \quad (46)$$

a horizon occurs at $x = x_\mathrm{h}$ satisfying

$$\beta(x_\mathrm{h}) = -s. \quad (47)$$

When the horizon lies within the range of the profile, its position is

$$x_\mathrm{h} = x_0 + \frac{1}{B}\,\mathrm{arctanh}\left(\frac{-s - C}{A}\right). \quad (48)$$

We denote the corresponding position in lattice units by

$$j_\mathrm{h} = \frac{x_\mathrm{h}}{\varepsilon} = \frac{L x_\mathrm{h}}{\ell}, \quad (49)$$

where j_h is a continuous coordinate in lattice units rather than an integer site index. For each geometry considered below, x_h and j_h refer to the horizon of the relevant branch. In particular, the black-hole horizon is associated with the a^- branch and satisfies $\beta(x_\mathrm{h}) = 1$, whereas the white-hole horizon is associated with the a^+ branch and satisfies $\beta(x_\mathrm{h}) = -1$.

Note that x_0 coincides with a horizon only when $|C| = 1$.

A. Black Hole

For the black-hole wave-packet dynamics shown in Figs. 2 and 3, we use $A = 0.6$, $B = 3$, $C = 0.6$, and $\ell = 2\pi$. In panels (a,b), we set $x_0 = 2\ell/3$, for which the event horizon where $\beta = 1$ lies at $j_h \approx 213$; in panels (c,d), we set $x_0 = \ell/3$, for which the event horizon lies at $j_h \approx 113$. The wave packets in Figs. 2(a,b) and 3(a,b) are initially placed outside the black hole, whereas those in Figs. 2(c,d) and 3(c,d) are initially placed inside the black hole.

We first examine the motion of the wave packet for the case $p = 0$. Fig. 2 shows the time evolution of wave packets for several initial conditions. The event horizon is indicated by the black dotted line, while the geodesic is shown by the light-blue curve. The region to the right of the event horizon corresponds to the black-hole interior. Figs. 2(a,b) show that wave packets initially placed outside the black hole follow the corresponding geodesic trajectories for both ingoing and outgoing motion. Figs. 2(c,d) show the time evolution of a wave packet initially placed inside the black hole. Inside the event horizon, both characteristic velocities are positive in this setup, so wave packets propagate only toward increasing x. Thus, the wave packet is expected to propagate only to the right. The numerical results show that both the right-moving mode in Fig. 2(c) and the left-moving mode in Fig. 2(d) propagate in the rightward direction. The initial state was chosen with $j_0 = 150$, and the standard deviation was set to $\sigma = 0.05L$.

To make weak-amplitude features visible without discarding their sign, we use a symmetric asinh mapping for the color scale in Fig. 3(a) and Fig. 6(a):

$$C(\mathcal{H}) = \frac{1}{2}\left[1 + \frac{\mathrm{asinh}(\mathcal{H}/\mathcal{H}_0)}{\mathrm{asinh}(\mathcal{H}_\mathrm{max}/\mathcal{H}_0)}\right], \qquad \mathcal{H}_\mathrm{max} = \max_{j,t}|\mathcal{H}_j(t)|. \quad (50)$$

Here, $C \in [0, 1]$ is the color coordinate and $\mathcal{H}_0$ sets the width of the approximately linear region around zero. This mapping is used only for visualization; the color-bar labels show the untransformed energy-density values.

Next, we investigate the motion of the wave packet for the case $p = 1$. As in the case $p = 0$, Figs. 3(a,b) show that wave packets initially placed outside the black hole follow the corresponding geodesic trajectories for both ingoing and outgoing motion. Figs. 3(c,d) show that wave packets initially placed inside the black hole again prop-

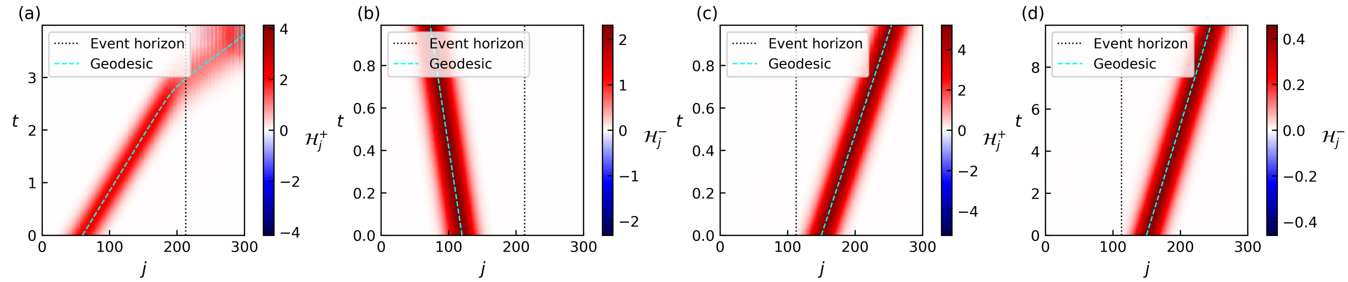

FIG. 2. Wave packet dynamics for the black-hole profile with $p = 0$. The system parameters are $L = 300$, $\ell = 2\pi$, and $\beta(x) = A\tanh[B(x - x_0)] + C$ with $A = 0.6$, $B = 3$, and $C = 0.6$. In (a,b), $x_0 = 2\ell/3$ and the event horizon is at $j_h \approx 213$; in (c,d), $x_0 = \ell/3$ and the event horizon is at $j_h \approx 113$. (a) Packet initialized outside the horizon with ingoing group velocity ($j_0 = 60$, $\sigma = 0.05L$). (b) Packet initialized outside the horizon with outgoing group velocity ($j_0 = 120$, $\sigma = 0.05L$). (c) Packet initialized inside the horizon with a right-moving initial spinor ($j_0 = 150$, $\sigma = 0.05L$). (d) Packet initialized inside the horizon with a left-moving initial spinor ($j_0 = 150$, $\sigma = 0.05L$). Black dotted lines indicate the event horizon, and cyan curves indicate the corresponding null geodesics $dx/dt = \beta(x) \pm 1$.

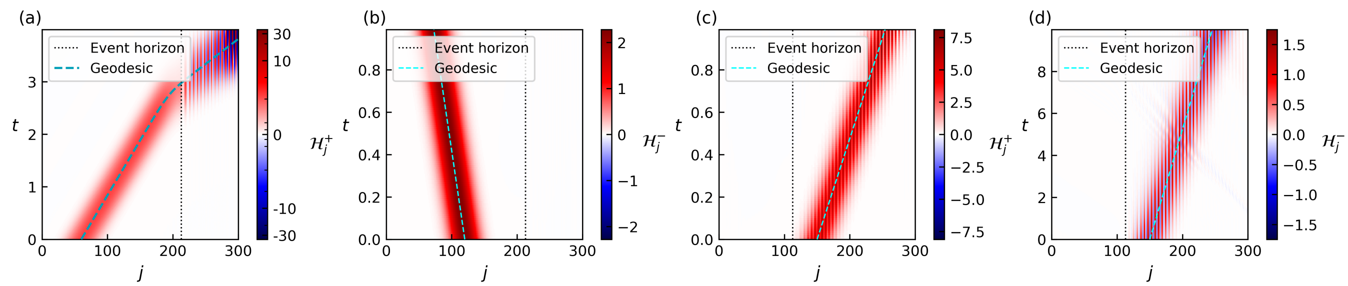

FIG. 3. Wave packet dynamics for the black-hole profile with $p = 1$. System parameters are the same as in Fig. 2: in (a,b), $x_0 = 2\ell/3$ and $j_h \approx 213$; in (c,d), $x_0 = \ell/3$ and $j_h \approx 113$. (a) Packet initialized outside the horizon with ingoing group velocity ($j_0 = 60$, $\sigma = 0.05L$). The symmetric asinh color mapping in Eq. (50) is used with $\mathcal{H}_0 = 1$. (b) Packet initialized outside the horizon with outgoing group velocity ($j_0 = 120$, $\sigma = 0.05L$). (c) Packet initialized inside the horizon with a right-moving initial spinor ($j_0 = 150$, $\sigma = 0.05L$). (d) Packet initialized inside the horizon with a left-moving initial spinor ($j_0 = 150$, $\sigma = 0.05L$). Black dotted lines indicate the event horizon, and cyan curves indicate the corresponding null geodesics $dx/dt = \beta(x) \pm 1$. Interior oscillations in (c,d) are more pronounced than in the $p = 0$ case owing to the stronger excitation of doubler modes, as confirmed by the k-space analysis of Fig. 4.

agate toward increasing x, as expected from the positive characteristic velocities inside the horizon. Inside the black hole, the wave packet exhibits pronounced oscillatory behavior, whose origin can be traced to the doubler modes. For the packet entering from the exterior [Fig. 3(a)], the oscillation develops after the packet reaches the interior. This is caused by the excitation of doubler modes during time evolution. Fig. 4(a) shows the low-energy doubler modes that appear for $p = 1$ inside the horizon, where $\beta > 1$, while Fig. 4(b) shows a secondary Fourier peak growing near the predicted wavenumber. On the other hand, the oscillations in Figs. 3(c,d) are already present from the initial time. Since these wave packets are initialized inside the black hole, they already have overlap with the doubler branch indicated in Fig. 4(a). Although the $p = 0$ model also contains a doubler branch near $k = \pi$, the FFT analysis shows a larger doubler-mode peak for $p = 1$ in the black-hole interior, explaining the more pronounced oscillations in Figs. 3(c,d).

We next extract the surface gravity of the effective horizon from the near-horizon wave-packet dynamics. For a left-moving wave packet localized in the exterior region near the event horizon, the linearized near-horizon null-geodesic equation gives the approximate growth of the standard deviation

$$\delta\sigma(t) \propto e^{\kappa t} - 1, \tag{51}$$

where $\kappa = |\beta'(x_h)|$ is the surface gravity. This approximate relation follows by linearizing the left-moving null trajectory $dx/dt = \beta(x) - 1$ near the horizon, where $\beta(x_h) = 1$, giving $d(x - x_h)/dt \simeq \kappa(x - x_h)$ and hence exponential stretching of the wave packet. It implies that the surface gravity can be read off from the exponential growth rate of $\delta\sigma$ as long as the wave packet remains in the near-horizon regime. Here, the wave-packet width $\sigma(t)$ is calculated as the standard deviation of the spatial profile. Numerically, the nonnegative weights are obtained from the absolute value of the vacuum-subtracted $\mathcal{H}_j^-$ profile, and the width in lattice-site units is converted

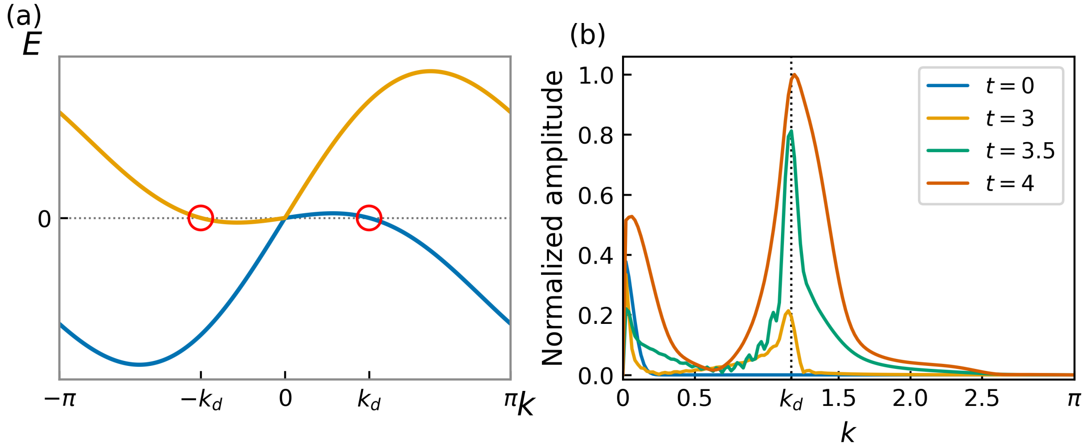

FIG. 4. Excitation of doubler modes in the oscillatory wave-packet dynamics for $p = 1$. (a) Local dispersion relation at $\beta = 1.2$, representative of the black-hole interior. The red circles mark the low-energy crossings at $k = \pm k_d$, corresponding to the doubler modes, where $k_d \simeq 1.171$. (b) Spatial FFT magnitude of the mean-subtracted branch-resolved profile $\delta\mathcal{E}_{j,+}$ in Fig. 3(a) at successive times $t = 0, 3, 3.5, 4$, normalized by the maximum over the four spectra. The black dotted line indicates the doubler-mode wave number k_d predicted from (a). The peak near $k = 0$ corresponds to the low-momentum quasiparticle component, while the secondary peak grows near k_d as the packet enters the black-hole interior. The secondary peak is already present before substantial boundary overlap, supporting its identification with doubler-mode excitation; its late-time amplitude at $t = 3.5$ and 4 can also be enhanced by contact with the right boundary.

to physical length by multiplication by $\varepsilon = \ell/L$. We define $\delta\sigma(t) = \sigma(t) - \sigma(0)$. In practice, we fit the numerical data to $a(e^{\kappa_{\rm fit} t} - 1)$, where a is a prefactor and $\kappa_{\rm fit}$ is the fitted growth rate.

For the surface-gravity extraction (Fig. 5), we use $A = 1$, $B = 1$, $C = 1$, $x_0 = \pi$ (so $x_h = x_0$, $\kappa = |\beta'(x_h)| = 1$, and $j_h = 0.5L$), $j_0 = 245$, $\sigma = 0.003L$, and fit the data over the time interval $t \in [0, 1]$. Fig. 5 compares the numerical data with the analytical near-horizon form for $\kappa = 1$, while the quoted $\kappa_{\rm fit}$ values are obtained by allowing the exponential growth rate to vary. As shown in Fig. 5(a), for $p = 0$, the fitted growth rate agrees with the analytical surface gravity $\kappa = 1$ within a relative error of approximately 3.3%. Small deviations from the analytical near-horizon curve at later times can be attributed to the breakdown of the approximation in Eq. (51) as the wave packet moves away from the event horizon.

In contrast, for $p = 1$, the discrepancy is larger than in the $p = 0$ case. The relative error of the fitted growth rate with respect to the analytical value is about 4.9%. This discrepancy can be attributed to deformation of the wave packet as it broadens, which causes deviations from the simple exponential behavior.

B. White Hole

In this section, we present the results of numerical simulations of particle dynamics in a white-hole spacetime. We consider the case in which a wave packet is incident from the exterior toward the interior of the white hole. We also discuss the finite-size effects that arise in this setup. For the white-hole dynamics, we use $A = C = -0.6$, $B = 3$, $x_0 = 2\ell/3$ (white-hole horizon at $j_h \approx 213$ for $L = 300$), with $j_0 = 0.2L$ and $\sigma = 0.05L$. For $p = 0$, as shown in Fig. 6, the wave packet is initially compressed at the white-hole horizon, yielding behavior consistent with the ideal analytical expectation. However, after a certain time, the wave packet collapses and is reflected. This behavior is attributed to a finite-size effect arising when the wave-packet standard deviation becomes smaller than the lattice spacing. For the stagnation-time scaling in Fig. 6(b), we use the same profile parameters, $A = C = -0.6$, $B = 3$, and $x_0 = 2\ell/3$, with $j_0 = 0.2L$ and $\sigma = 0.05L$. Near the white-hole horizon, the same linearized null-geodesic argument gives an exponential compression of the wave-packet standard deviation, $\sigma(t) \propto \sigma_0 e^{-\kappa t}$. In the lattice model, this continuum description breaks down once this standard deviation becomes comparable to the lattice spacing $\varepsilon = \ell/L$. Defining the continuum estimate of the stagnation time $T_{\rm cont}$ by $\sigma(T_{\rm cont}) \sim \varepsilon$, one obtains

$$T_{\rm cont} \propto \ln L, \tag{52}$$

for fixed continuum size ℓ and fixed initial physical standard deviation σ_0.

We then test this logarithmic scaling by examining the dependence of the lattice stagnation time $T_{\rm lat}$ on the system size L. To define $T_{\rm lat}$ in the numerical data, we identify an onset time $t_{\rm in}$ for the compression regime. Operationally, after the wave packet has entered the compression stage, $t_{\rm in}$ is taken as the first time at which $|\partial_x^2 \beta(\bar{x}(t))| < 0.1$, where $\bar{x}(t)$ is the wave-packet mean position. We then identify the time $t_{\rm min}$ at which the wave-

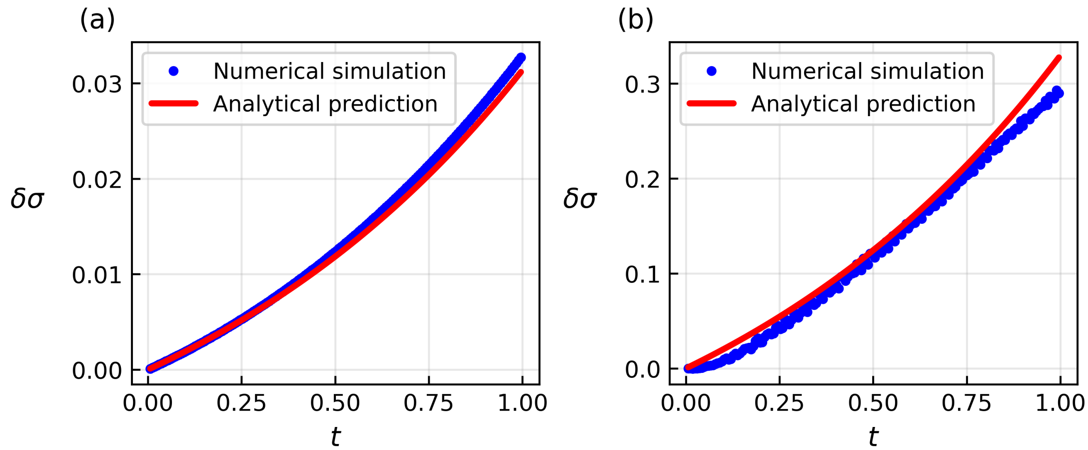

FIG. 5. Extraction of surface gravity from wave-packet broadening ($L = 500$, $A = B = C = 1$, $x_0 = \ell/2$, $j_0 = 245$, $\sigma = 0.003L$, with the fit performed over $t \in [0, 1]$). Blue dots show the numerical simulation results, and red curves show the analytical near-horizon scaling $a(e^{\kappa t} - 1)$ with $\kappa = 1$; the prefactor a fixes the vertical scale. The fitted growth rates are obtained separately from $a(e^{\kappa_{\rm fit} t} - 1)$. (a) Result for $p = 0$, with $|\kappa_{\rm fit} - \kappa|/\kappa \approx 3.3\%$. (b) Result for $p = 1$, with $|\kappa_{\rm fit} - \kappa|/\kappa \approx 4.9\%$ due to simultaneous wave packet dispersion. These percentages quantify the error in the fitted surface gravity, not the pointwise deviation of the red analytical curves. The vertical axis $\delta\sigma \equiv \sigma(t) - \sigma(0)$ denotes the change in the wave-packet standard deviation.

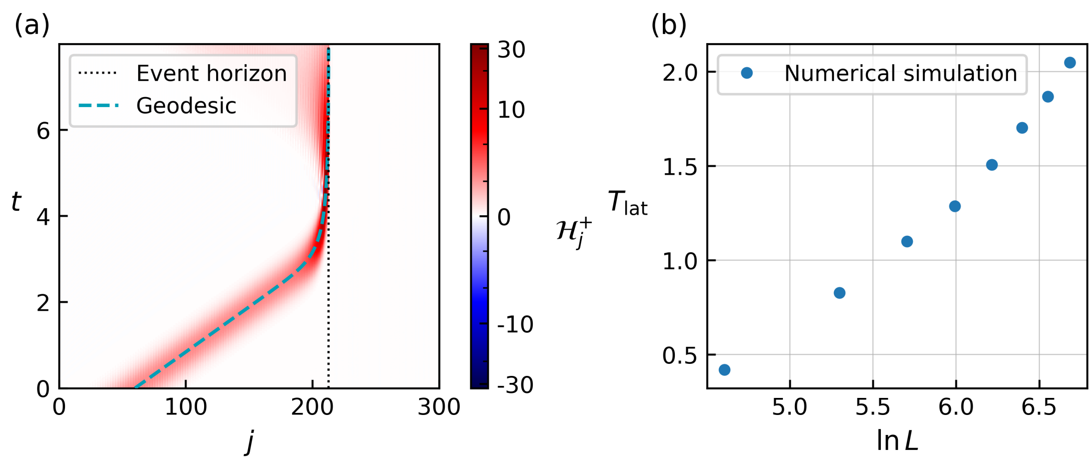

FIG. 6. Wave packet dynamics in the white-hole geometry for $p = 0$. (a) Space-time diagram of $\mathcal{H}_j^+$ for $L = 300$, $j_0 = 0.2L$, and $\sigma = 0.05L$, with β profile parameters $A = C = -0.6$, $B = 3$, and $x_0 = 2\ell/3$ (white-hole horizon at $j_h \approx 213$). The symmetric asinh color mapping in Eq. (50) is used with $\mathcal{H}_0 = 3$. The wave packet is compressed toward the white-hole horizon (black dotted line) and eventually reflected due to finite lattice resolution. (b) Lattice stagnation time $T_{\rm lat}$ versus $\ln L$ for various system sizes, using $A = C = -0.6$, $B = 3$, $x_0 = 2\ell/3$, $j_0 = 0.2L$, and $\sigma = 0.05L$. Blue dots show numerical simulation results; their approximately linear dependence on $\ln L$ is consistent with the logarithmic scaling of Eq. (52).

packet standard deviation reaches its minimum. The lattice stagnation time is then defined operationally as $T_{\rm lat} = t_{\rm min} - t_{\rm in}$. The approximately linear dependence of the data on $\ln L$ in Fig. 6(b) is consistent with the logarithmic system-size scaling predicted for $T_{\rm cont}$. This comparison tests only the logarithmic dependence, not its coefficient: $T_{\rm lat}$ is defined operationally from the nonlinear lattice collapse and reflection, so its slope need not equal the continuum value $1/\kappa$. Therefore, we conclude that the reflection of the wave packet is caused by a finite-size effect associated with its standard deviation becoming smaller than the lattice spacing.

For $p = 1$, the wave-packet dynamics differs from the $p = 0$ case. As shown in Fig. 7, part of the wave packet crosses the white-hole horizon instead of being simply reflected. This behavior is absent for $p = 0$, indicating that the coupling between the right- and left-moving modes plays an important role.

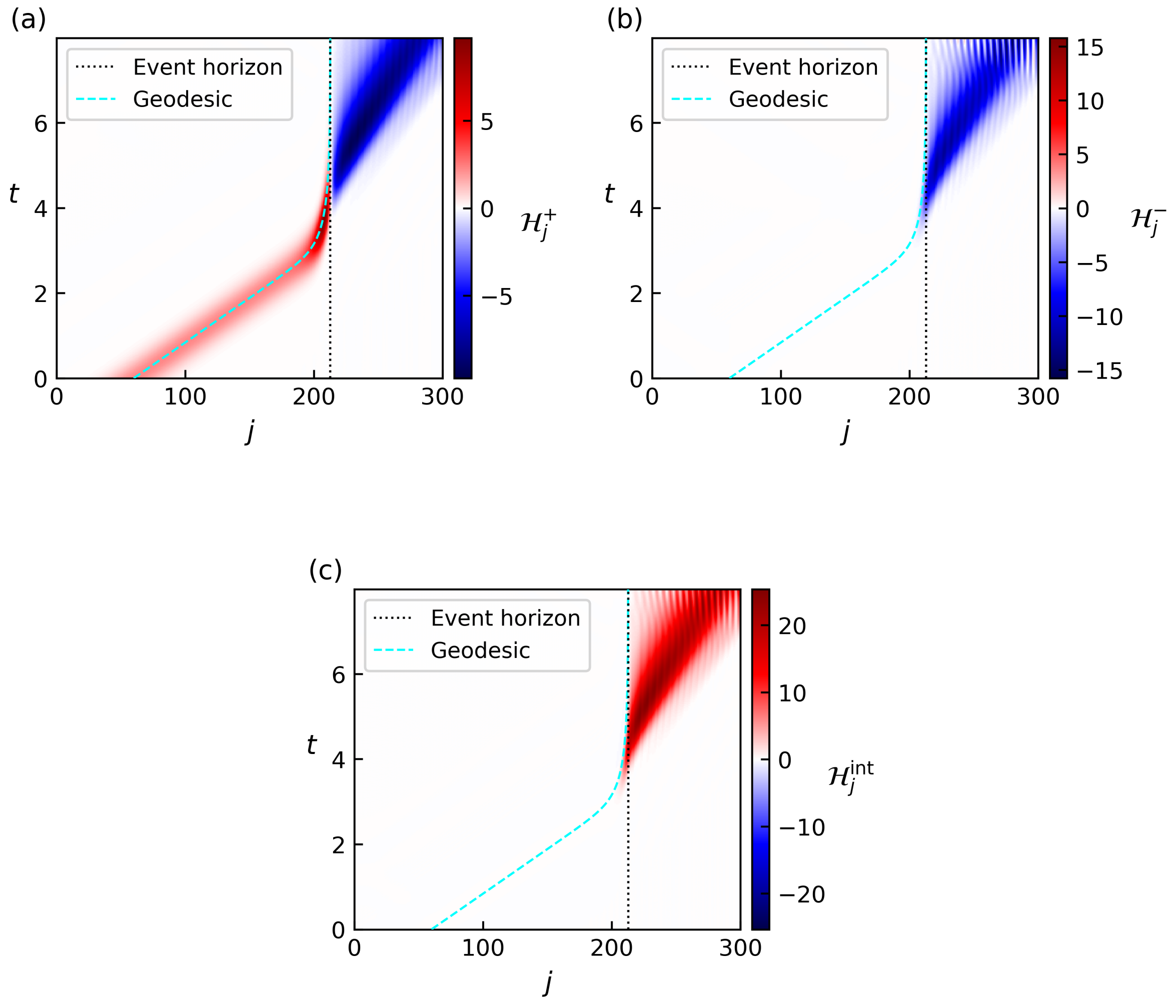

FIG. 7. White-hole dynamics for $p = 1$ ($L = 300$, $j_0 = 0.2L$, $\sigma = 0.05L$, with the same β profile as Fig. 6(a), $A = C = -0.6$, $B = 3$, $x_0 = 2\ell/3$, and $j_h \approx 213$). (a) Space-time diagram of $\mathcal{H}_j^+$. (b) Space-time diagram of $\mathcal{H}_j^-$. (c) Space-time diagram of the interaction contribution $\mathcal{H}_j^{\rm int}$. Unlike the $p = 0$ case, the coupling between the right- and left-moving components allows partial transmission across the white-hole horizon (black dotted line).

IV. CONCLUSIONS

In this work, we have investigated particle dynamics in a one-dimensional lattice fermion model designed to simulate black-hole and white-hole spacetimes, focusing on the correspondence between lattice wave-packet trajectories and geodesics of the effective curved metric.

For the black-hole geometry, Gaussian wave packets initialized both outside and inside the event horizon closely follow the corresponding geodesics for both $p = 0$ and $p = 1$ parameter choices. The oscillatory behavior observed inside the black hole arises from doubler modes, with the $p = 1$ case exhibiting more pronounced oscillations owing to stronger overlap with, and excitation of, these modes. We also extracted the surface gravity from the exponential growth of the wave-packet standard deviation in the near-horizon regime, finding agreement within a relative error of $\sim 3.3\%$ for $p = 0$ and $\sim 4.9\%$ for $p = 1$, with the remaining discrepancies attributed to deformation of the wave packet during broadening.

For the white-hole geometry, the wave packet is compressed toward the white-hole horizon in a manner consistent with the continuum expectation, but eventually collapses and reflects due to a finite-size effect. The lattice stagnation time is consistent with the logarithmic system-size scaling $T_{\rm lat} \propto \ln L$. For $p = 1$, partial transmission across the white-hole horizon is observed, indicating that the coupling between left- and right-moving modes plays an important role.

These results demonstrate that the one-dimensional lattice fermion model provides a controlled setting for assessing curved-spacetime particle dynamics in lattice fermion simulators. Future directions include extending the model to dynamical black-hole backgrounds to investigate Hawking radiation from time-dependent horizons, as well as implementing the lattice Hamiltonian on near-term quantum hardware to experimentally probe this effect.